Major Project (BTCS706N) - Weekly Progress Report

Project Title: Quantum Secure Email Client Application (QuMail)

Team ID: G14

Reporting Period: Week 5

Phase: Phase 2 – Object-Oriented Analysis

#Executive Summary

Current Sprint: Sprint 5 of 6

Sprint Goal: Continue Phase 2 – Object-Oriented Analysis with a focus on Interaction Modeling. The objective is to represent the interactions and communication between system objects over time using a UML Sequence Diagram for the secure email communication workflow.

Status: In Progress

During Week 5, the interaction workflow of the Quantum Secure Email Client Application was modelled using a UML Sequence Diagram. The diagram illustrates how different system components communicate during user authentication, email composition, quantum key generation, AES encryption, secure email transmission through the SMTP server, email reception, and decryption.

The Week 3 Object Diagram, Initial Class Diagram, and Week 4 Activity Diagram serve as references for developing the interaction model.

#UML & OOAD Progress

Diagram Prepared: UML Sequence Diagram – represents the interactions between the sender, email client interface, authentication module, Quantum Key Manager, AES Encryption Module, SMTP Server, and receiver over time.

Earlier Diagrams Referenced:

- Object Diagram – Represents specific object instances and their relationships within the system.
- Initial Class Diagram (Domain Model) – Identifies the main domain classes, attributes, and relationships.
- Activity Diagram – Represents the dynamic workflow of secure email communication, including authentication, encryption, transmission, and decryption.

OO Concept Applied: Interaction Modeling focuses on communication between objects through messages over time. The Sequence Diagram shows the order of operations, message exchanges, and responsibilities of different components during secure email transmission.

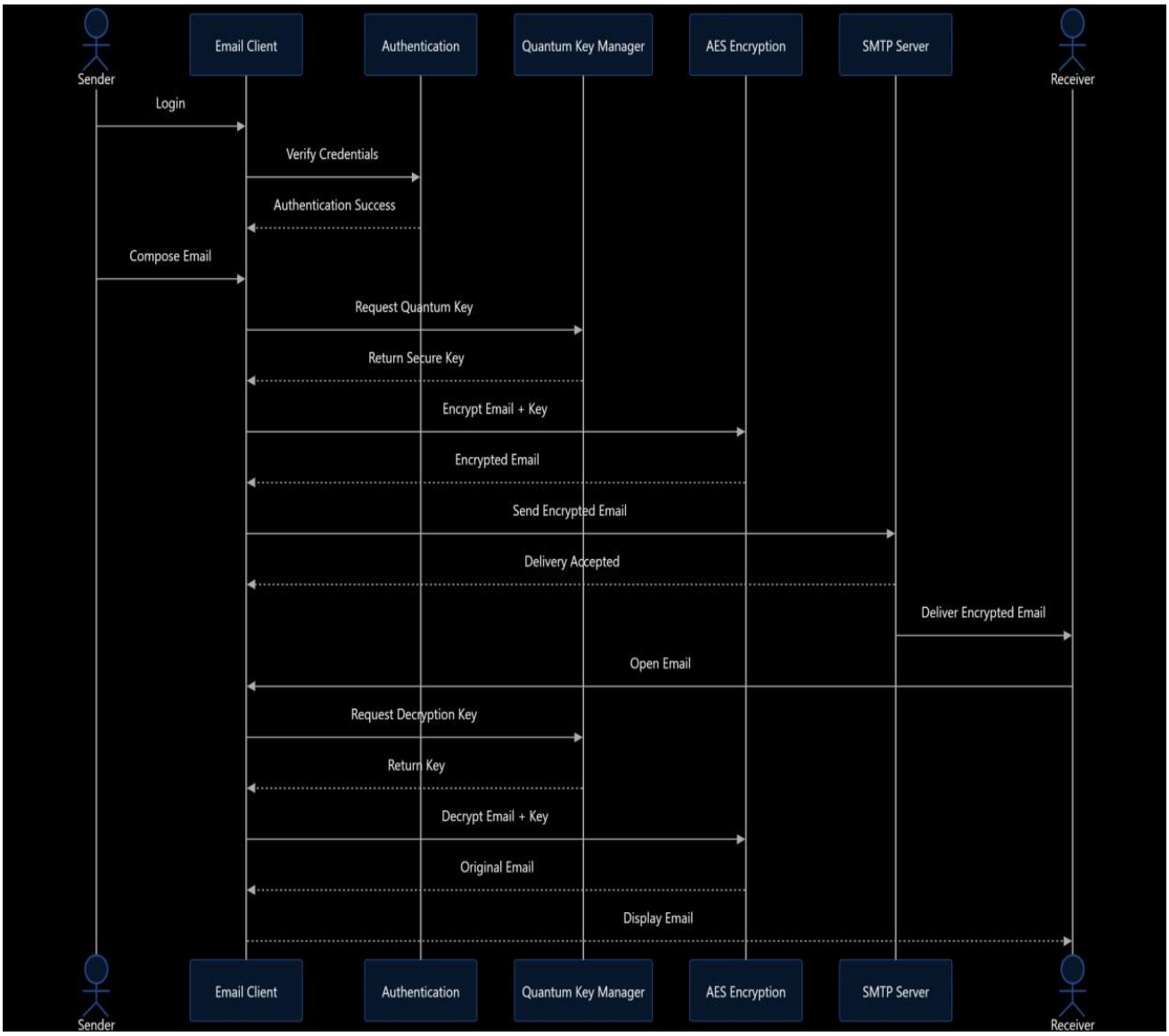
#Sequence Diagram Overview

The Sequence Diagram represents the complete secure email communication process:

1. The sender opens the QuMail application and logs in.
2. The Authentication Module verifies the user's credentials.
3. The sender composes an email using the client interface.
4. The client requests a secure encryption key from the Quantum Key Manager.
5. The AES Encryption Module encrypts the email content.
6. The encrypted email is sent to the SMTP Server.
7. The SMTP Server delivers the encrypted email to the receiver.
8. The receiver accesses the received email.
9. The system obtains the required decryption key.
10. The AES module decrypts the email and displays the original message to the receiver.

The Sequence Diagram demonstrates the ordered communication between system components during secure email transmission.

UML Deliverable – Sequence Diagram



#Technical Execution & Testing Summary

Technical Execution (DevOps & Git)

Jira / Project Tasks:

- Interaction-modeling activities for Phase 2 were addressed through the secure-email communication workflow.
- System objects and participating components were identified for the Sequence Diagram.
- Messages and interactions between the sender, client interface, authentication module, Quantum Key Manager, AES module, SMTP Server, and receiver were identified.
- The UML Sequence Diagram was prepared for inclusion in the Week 5 project documentation.
- Jira task: SCRUM-21 (Sequence Diagram).

Git Activity:

- The Sequence Diagram and Week 5 report were saved in the project folder.
- The documentation changes were committed using VS Code Source Control.
- The changes were pushed to the GitHub repository.

CI/CD Status: CI/CD implementation is not the primary focus of this report. Week 5 focuses on Interaction Modeling and UML Sequence Diagram development.

Testing Summary

Total Unit Tests: 0

Pass Rate: N/A

Validation: Review the sequence of messages, object interactions, communication order, and consistency between the Sequence Diagram and earlier UML models.

Key Bug Identified: None. This week focused on UML modelling; no implementation bugs were observed.

#Goals for Next Week

- Review and refine the Sequence Diagram based on project-guide and mentor feedback.
- Check consistency between the Week 3 static models, Week 4 Activity Diagram, and Week 5 Sequence Diagram.
- Continue the remaining Phase 2 project activities according to the approved project plan.
- Maintain UML documentation and project changes through Git.

Mentor Signature: _____ Project Guide Signature: _____